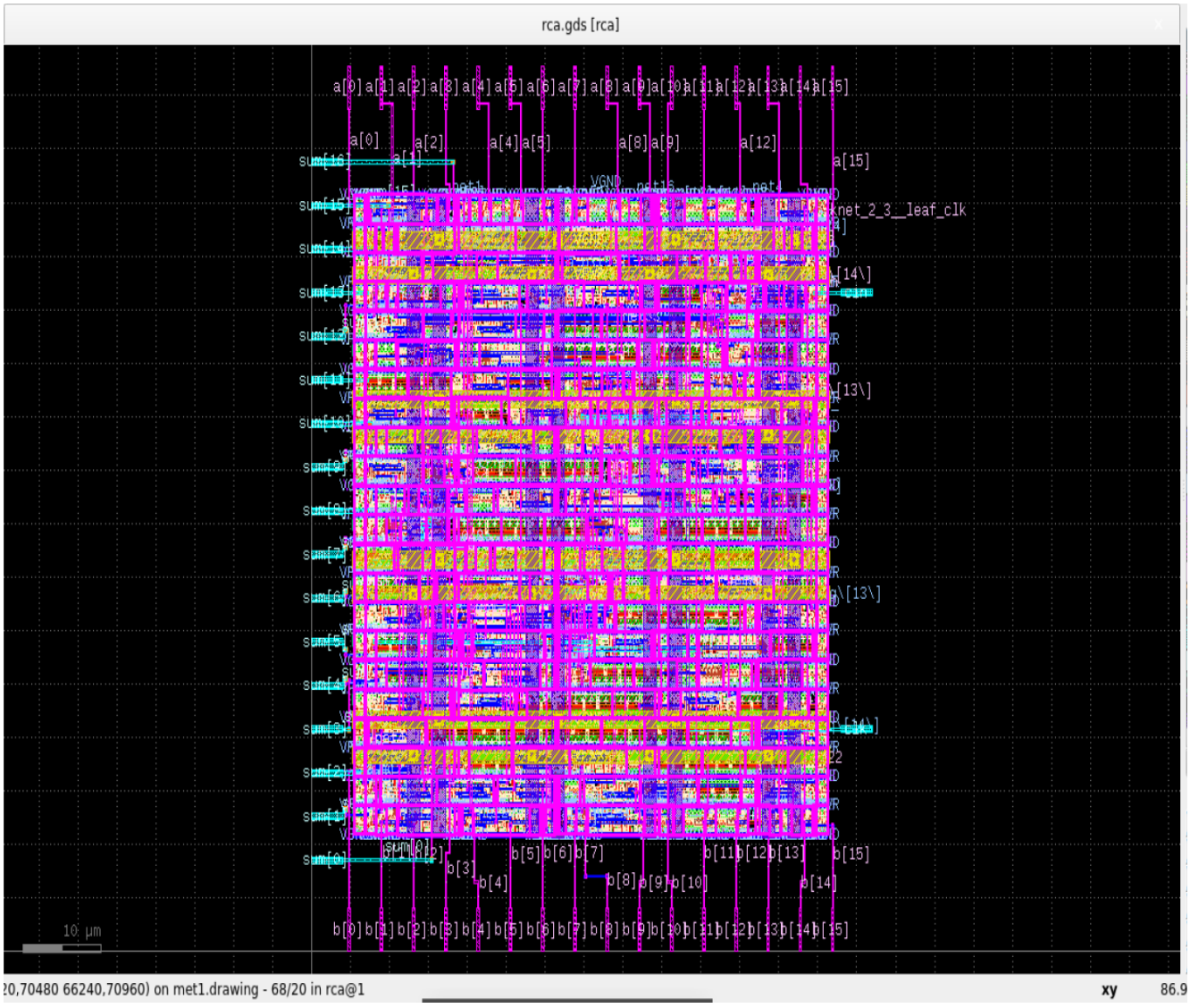
EE5311: Digital IC Design
Tutorial 10

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Roll Number: EE22B032

October 27, 2025

Ripple Carry Adder

Layout



Code

```
    module rca (
    output reg [16:0] sum,   // 16-bit sum + 1-bit final carry-out
    input  [15:0]     a,
    input  [15:0]     b,
    input             cin,
    input             clk
);

    logic stage
    logic [15:0] a_q, b_q;
    logic        cin_q;

    always @(posedge clk) begin
        a_q   <= a;
        b_q   <= b;
        cin_q <= cin;
    end


    logic [16:0] carry;
    logic [15:0] sum_comb;
    logic [16:0] sum_d;


    assign carry[0] = cin_q;

    genvar i;
    generate
        for (i = 0; i < 16; i = i + 1) begin : rca_full_adder
          assign sum_comb[i] = a_q[i] ^ b_q[i] ^ carry[i];
          assign carry[i+1]  = (a_q[i] & b_q[i]) | (a_q[i] & carry[i]) | (b_q[i] & carry[i])
        end
    endgenerate

    assign sum_d = {carry[16], sum_comb};

    always @(posedge clk) begin
        sum <= sum_d;
    end

endmodule
```

Report

```
40218 skew_report
40219
40220 ========================================================================
40221 report_clock_skew
40222 ========================================================================
40223 Clock clock
40224 Latency      CRPR       Skew
40225 _179_/CLK ^
40226    0.50
40227 _166_/CLK ^
40228    0.49      0.00       0.01
40229
40230 skew_report_end
40231 summary_report
40232
40233 ========================================================================
40234 report_tns
40235 ========================================================================
40236 tns 0.00
40237
40238 ========================================================================
40239 report_wns
40240 ========================================================================
40241 wns 0.00
40242
40243 ========================================================================
40244 report_worst_slack -max (Setup)
40245 ========================================================================
40246 worst slack 0.05
40247
40248 ========================================================================
40249 report_worst_slack -min (Hold)
40250 ========================================================================
40251 worst slack 0.37
40252 summary_report_end
40253 check_nonpropagated_clocks
40254 check_nonpropagated_clocks_end
40255 [WARNING] Did not save OpenROAD database!
40256 Writing SDF files for all corners…
40257 Writing SDF for the Fastest corner to /openlane/designs/rca/runs/auto_pnr/results/routing/mca/process_corner_nom/rca.Fastest.sdf…
40258 Writing SDF for the Slowest corner to /openlane/designs/rca/runs/auto_pnr/results/routing/mca/process_corner_nom/rca.Slowest.sdf…
40259 Writing SDF for the Typical corner to /openlane/designs/rca/runs/auto_pnr/results/routing/mca/process_corner_nom/rca.Typical.sdf…
40260 Writing timing models for all corners…
40261 Writing timing models for the Fastest corner to /openlane/designs/rca/runs/auto_pnr/results/routing/mca/process_corner_nom/rca.Fastest.lib…
```

$$T_{\min} = \text{Target Period} - \text{Worst Setup Slack}$$

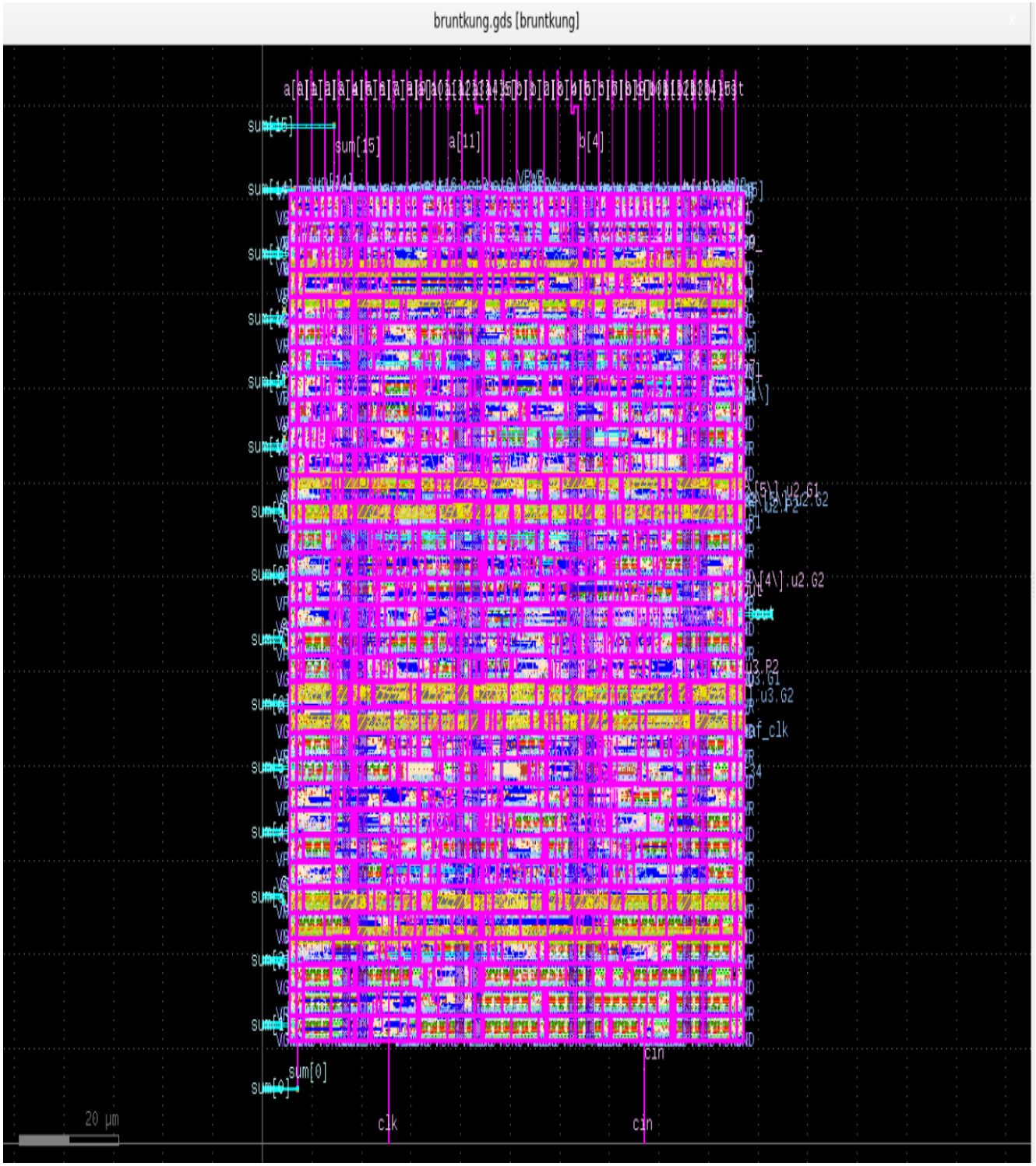
$$T_{\min} = 6.0 \text{ ns} - 0.05 \text{ ns} = \mathbf{5.95 \text{ ns}}$$

$$F_{\max} = \frac{1}{T_{\min}}$$

$$F_{\max} = \frac{1}{5.95 \text{ ns}} \approx \mathbf{168.07 \text{ MHz}}$$

Bruntkung Adder

Layout



Code

```
module bruntkung #(
  parameter WIDTH = 16
)(
  input  wire                 clk,
  input  wire                 rst,
  input  wire [WIDTH-1:0]     a,
  input  wire [WIDTH-1:0]     b,
  input  wire                 cin,
  output reg  [WIDTH-1:0]     sum,
  output reg                  cout
);

  // STAGE 0: Register Inputs
  reg [WIDTH-1:0] a_r, b_r;
  reg             cin_r;
  always @(posedge clk) begin
    if (rst) begin
      a_r   <= 0;
      b_r   <= 0;
      cin_r <= 0;
    end else begin
      a_r   <= a;
      b_r   <= b;
      cin_r <= cin;
    end
  end

  // STAGE 1: Compute bitwise P/G

  wire [WIDTH-1:0] P1, G1;
  genvar i;
  generate
    for (i = 0; i < WIDTH; i = i + 1) begin : PG1
      assign P1[i] = a_r[i] ^ b_r[i];
      assign G1[i] = a_r[i] & b_r[i];
    end
  endgenerate

  reg [WIDTH-1:0] P1_r, G1_r;
  always @(posedge clk) begin
    if (rst)      {P1_r, G1_r} <= 0;
    else          {P1_r, G1_r} <= {P1,  G1 };
  end

  // STAGE 2: Level-2 prefix (pairs)

  wire [WIDTH/2-1:0] P2, G2;
  generate
    for (i = 0; i < WIDTH/2; i = i + 1) begin : LVL2
      pg_dot u2(
```

```
      .P1  (P1_r[2*i]),   .G1(G1_r[2*i]),
      .P2  (P1_r[2*i+1]), .G2(G1_r[2*i+1]),
      .Pout(P2[i]),       .Gout(G2[i])
    );
  end
endgenerate

reg [WIDTH/2-1:0] P2_r, G2_r;
always @(posedge clk) begin
  if (rst)      {P2_r, G2_r} <= 0;
  else          {P2_r, G2_r} <= {P2,  G2 };
end

// STAGE 3: Level-3 prefix (groups of 4)

wire [WIDTH/4-1:0] P3, G3;
generate
  for (i = 0; i < WIDTH/4; i = i + 1) begin : LVL3
    pg_dot u3(
      .P1  (P2_r[2*i]),   .G1(G2_r[2*i]),
      .P2  (P2_r[2*i+1]), .G2(G2_r[2*i+1]),
      .Pout(P3[i]),       .Gout(G3[i])
    );
  end
endgenerate

reg [WIDTH/4-1:0] P3_r, G3_r;
always @(posedge clk) begin
  if (rst)      {P3_r, G3_r} <= 0;
  else          {P3_r, G3_r} <= {P3,  G3 };
end

// STAGE 4: Level-4 prefix (groups of 8)

wire [WIDTH/8-1:0] P4, G4;
generate
  for (i = 0; i < WIDTH/8; i = i + 1) begin : LVL4
    pg_dot u4(
      .P1  (P3_r[2*i]),   .G1(G3_r[2*i]),
      .P2  (P3_r[2*i+1]), .G2(G3_r[2*i+1]),
      .Pout(P4[i]),       .Gout(G4[i])
    );
  end
endgenerate

reg [WIDTH/8-1:0] P4_r, G4_r;
always @(posedge clk) begin
  if (rst)      {P4_r, G4_r} <= 0;
  else          {P4_r, G4_r} <= {P4,  G4 };
end
```

```
  // STAGE 5: Level-5 prefix (entire 16)

  wire P5, G5;
  pg_dot u5(
    .P1  (P4_r[0]), .G1(G4_r[0]),
    .P2  (P4_r[1]), .G2(G4_r[1]),
    .Pout(P5),      .Gout(G5)
  );

  reg P5_r, G5_r;
  always @(posedge clk) begin
    if (rst)      {P5_r, G5_r} <= 0;
    else          {P5_r, G5_r} <= {P5,  G5 };
  end

  // STAGE 6: Compute final carry chain and register sum & cout

  wire [WIDTH:0] carry;
  assign carry[0]  = cin_r;
  assign carry[1]  = G1_r[0]     | (P1_r[0]     & carry[0]);
  assign carry[2]  = G2_r[0]     | (P2_r[0]     & carry[0]);
  assign carry[3]  = G2_r[1]     | (P2_r[1]     & carry[1]);
  assign carry[4]  = G3_r[0]     | (P3_r[0]     & carry[0]);
  assign carry[5]  = G3_r[1]     | (P3_r[1]     & carry[1]);
  assign carry[6]  = G3_r[2]     | (P3_r[2]     & carry[2]);
  assign carry[7]  = G3_r[3]     | (P3_r[3]     & carry[3]);
  assign carry[8]  = G4_r[0]     | (P4_r[0]     & carry[0]);
  assign carry[9]  = G1_r[8]     | (P1_r[8]     & carry[8]);
  assign carry[10] = G2_r[4]     | (P2_r[4]     & carry[8]);
  assign carry[11] = G1_r[10]    | (P1_r[10]    & carry[10]);
  assign carry[12] = G3_r[2]     | (P3_r[2]     & carry[8]);
  assign carry[13] = G1_r[12]    | (P1_r[12]    & carry[12]);
  assign carry[14] = G2_r[6]     | (P2_r[6]     & carry[12]);
  assign carry[15] = G1_r[14]    | (P1_r[14]    & carry[14]);
  assign carry[16] = G5_r        | (P5_r        & carry[0]);

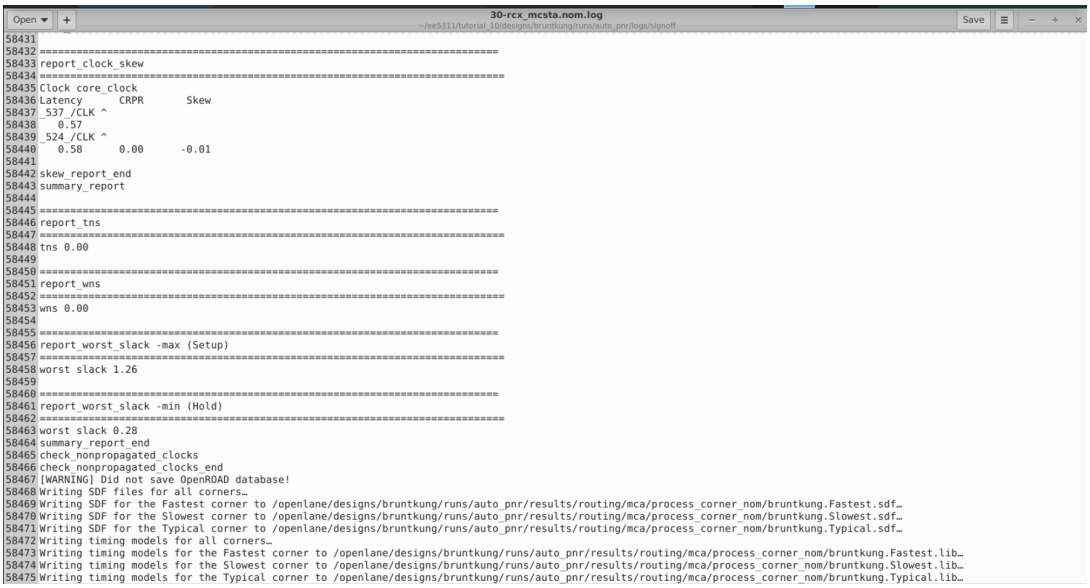
  always @(posedge clk) begin
    if (rst) begin
      sum  <= 0;
      cout <= 0;
    end else begin
      sum  <= P1_r ^ carry[WIDTH-1:0];
      cout <= carry[WIDTH];
    end
  end

endmodule


module pg_dot (
```

```
  input  wire P1, G1,
  input  wire P2, G2,
  output wire Pout, Gout
);
  assign Gout = G2 | (P2 & G1);
  assign Pout = P2 & P1;
endmodule
```

Report



```
58431
58432 ===========================================================================
58433 report_clock_skew
58434 ===========================================================================
58435 Clock core_clock
58436 Latency       CRPR       Skew
58437 _537_/CLK ^
58438    0.57
58439 _524_/CLK ^
58440    0.58      0.00      -0.01
58441
58442 skew_report_end
58443 summary_report
58444
58445 ===========================================================================
58446 report_tns
58447 ===========================================================================
58448 tns 0.00
58449
58450 ===========================================================================
58451 report_wns
58452 ===========================================================================
58453 wns 0.00
58454
58455 ===========================================================================
58456 report_worst_slack -max (Setup)
58457 ===========================================================================
58458 worst slack 1.26
58459
58460 ===========================================================================
58461 report_worst_slack -min (Hold)
58462 ===========================================================================
58463 worst slack 0.28
58464 summary_report_end
58465 check_nonpropagated_clocks
58466 check_nonpropagated_clocks_end
58467 [WARNING] Did not save OpenROAD database!
58468 Writing SDF files for all corners…
58469 Writing SDF for the Fastest corner to /openlane/designs/bruntkung/runs/auto_pnr/results/routing/mca/process_corner_nom/bruntkung.Fastest.sdf…
58470 Writing SDF for the Slowest corner to /openlane/designs/bruntkung/runs/auto_pnr/results/routing/mca/process_corner_nom/bruntkung.Slowest.sdf…
58471 Writing SDF for the Typical corner to /openlane/designs/bruntkung/runs/auto_pnr/results/routing/mca/process_corner_nom/bruntkung.Typical.sdf…
58472 Writing timing models for all corners…
58473 Writing timing models for the Fastest corner to /openlane/designs/bruntkung/runs/auto_pnr/results/routing/mca/process_corner_nom/bruntkung.Fastest.lib…
58474 Writing timing models for the Slowest corner to /openlane/designs/bruntkung/runs/auto_pnr/results/routing/mca/process_corner_nom/bruntkung.Slowest.lib…
58475 Writing timing models for the Typical corner to /openlane/designs/bruntkung/runs/auto_pnr/results/routing/mca/process_corner_nom/bruntkung.Typical.lib…
```

$$T_{\min} = \text{Target Period} - \text{Worst Setup Slack}$$

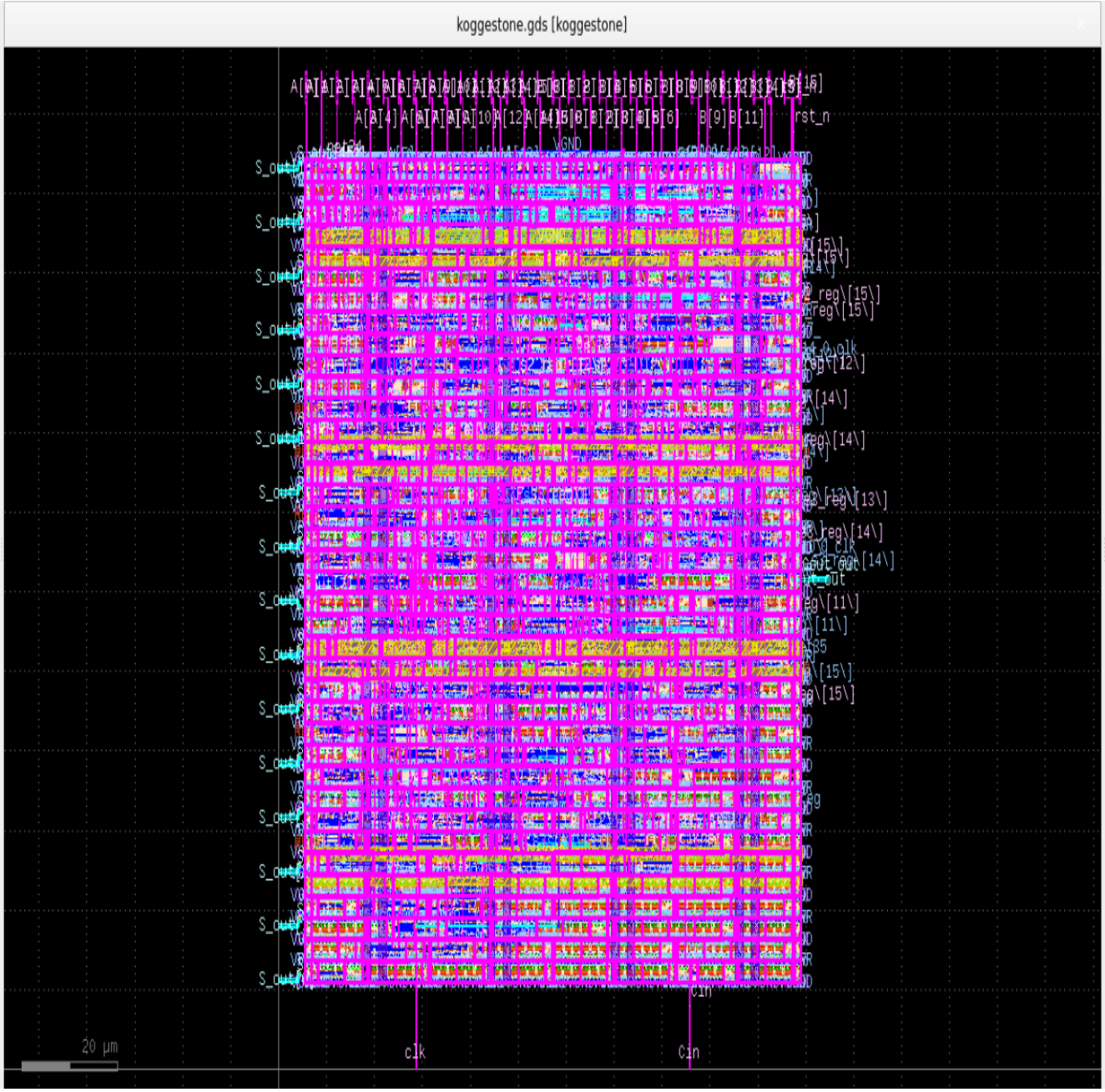
$$T_{\min} = 5.0 \text{ ns} - 1.26 \text{ ns} = \mathbf{3.74 \text{ ns}}$$

$$F_{\max} = \frac{1}{T_{\min}}$$

$$F_{\max} = \frac{1}{3.74 \text{ ns}} \approx \mathbf{267.38 \text{ MHz}}$$

Koggestone Adder

Layout



Code

```
module koggestone (
    input wire          clk,
    input wire          rst_n,

    input wire [15:0]   A,
    input wire [15:0]   B,
    input wire          Cin,

    output reg [15:0]   S_out,
    output reg          Cout_out
);

    localparam WIDTH = 16;

    // STAGE 1: G/P Generation

    wire [WIDTH-1:0] G_s1, P_s1;
    assign G_s1 = A & B;
    assign P_s1 = A ^ B;

    reg [WIDTH-1:0] G_s1_reg, P_s1_reg;
    reg Cin_s1_reg;

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            G_s1_reg   <= 0;
            P_s1_reg   <= 0;
            Cin_s1_reg <= 0;
        end else begin
            G_s1_reg   <= G_s1;
            P_s1_reg   <= P_s1;
            Cin_s1_reg <= Cin; // Cin is registered
        end
    end

    // STAGE 2: Prefix Tree (Levels 1 & 2)

    // Wires for logic in Stage 2
    wire [WIDTH-1:0] G1, P1; // Level 1 (distance 1)
    wire [WIDTH-1:0] G2, P2; // Level 2 (distance 2)

    genvar i;

    // Level 1 (dist 1)
    generate
        assign G1[0] = G_s1_reg[0];
        assign P1[0] = P_s1_reg[0];
        for (i = 1; i < WIDTH; i = i + 1) begin : gen_level1
            assign G1[i] = G_s1_reg[i] | (P_s1_reg[i] & G_s1_reg[i-1]);
            assign P1[i] = P_s1_reg[i] & P_s1_reg[i-1];
```

```
    end
endgenerate

// Level 2 (dist 2)
generate
    for (i = 0; i < 2; i = i + 1) begin : gen_level2_copy
        assign G2[i] = G1[i]; // Copy bits < distance
        assign P2[i] = P1[i];
    end
    for (i = 2; i < WIDTH; i = i + 1) begin : gen_level2
        assign G2[i] = G1[i] | (P1[i] & G1[i-2]);
        assign P2[i] = P1[i] & P1[i-2];
    end
endgenerate

reg [WIDTH-1:0] G2_s2_reg, P2_s2_reg;
reg [WIDTH-1:0] P_s2_reg;   // Pass-through P for final sum
reg             Cin_s2_reg; // Pass-through Cin

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        G2_s2_reg  <= 0;
        P2_s2_reg  <= 0;
        P_s2_reg   <= 0;
        Cin_s2_reg <= 0;
    end else begin
        G2_s2_reg  <= G2;
        P2_s2_reg  <= P2;
        P_s2_reg   <= P_s1_reg;   // Pass P from S1
        Cin_s2_reg <= Cin_s1_reg; // Pass Cin from S1
    end
end

// STAGE 3: Prefix Tree (Levels 3 & 4)

wire [WIDTH-1:0] G3, P3; // Level 3 (distance 4)
wire [WIDTH-1:0] G4, P4; // Level 4 (distance 8)

generate
    for (i = 0; i < 4; i = i + 1) begin : gen_level3_copy
        assign G3[i] = G2_s2_reg[i]; // Copy bits < distance
        assign P3[i] = P2_s2_reg[i];
    end
    for (i = 4; i < WIDTH; i = i + 1) begin : gen_level3
        assign G3[i] = G2_s2_reg[i] | (P2_s2_reg[i] & G2_s2_reg[i-4]);
        assign P3[i] = P2_s2_reg[i] & P2_s2_reg[i-4];
    end
endgenerate

generate
    for (i = 0; i < 8; i = i + 1) begin : gen_level4_copy
```

```
        assign G4[i] = G3[i]; // Copy bits < distance
        assign P4[i] = P3[i];
    end
    for (i = 8; i < WIDTH; i = i + 1) begin : gen_level4
        assign G4[i] = G3[i] | (P3[i] & G3[i-8]);
        assign P4[i] = P3[i] & P3[i-8];
    end
endgenerate

reg [WIDTH-1:0] G4_s3_reg, P4_s3_reg;
reg [WIDTH-1:0] P_s3_reg;   // Pass-through P for final sum
reg             Cin_s3_reg; // Pass-through Cin

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        G4_s3_reg  <= 0;
        P4_s3_reg  <= 0;
        P_s3_reg   <= 0;
        Cin_s3_reg <= 0;
    end else begin
        G4_s3_reg  <= G4;
        P4_s3_reg  <= P4;
        P_s3_reg   <= P_s2_reg;   // Pass P from S2
        Cin_s3_reg <= Cin_s2_reg; // Pass Cin from S2
    end
end

// STAGE 4: Final Carry and Sum Calculation

wire [WIDTH-1:0] C_in; // Carry-in array for each bit
wire [WIDTH-1:0] S_s4;
wire Cout_s4;

assign C_in[0] = Cin_s3_reg;
generate
    for (i = 1; i < WIDTH; i = i + 1) begin : gen_carry
        assign C_in[i] = G4_s3_reg[i-1] | (P4_s3_reg[i-1] & Cin_s3_reg);
    end
endgenerate

assign S_s4 = P_s3_reg ^ C_in;

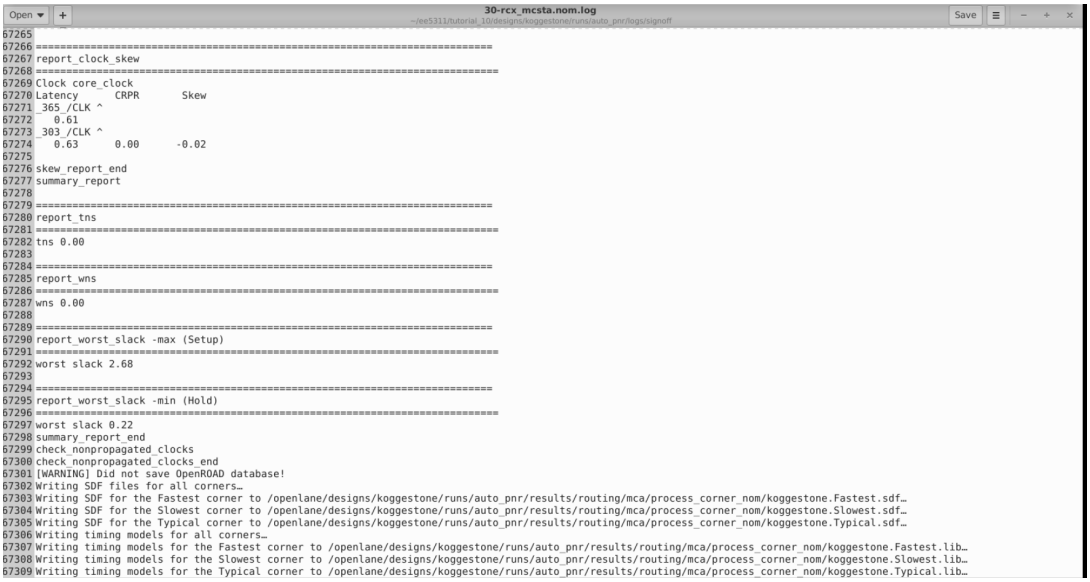
assign Cout_s4 = G4_s3_reg[WIDTH-1] | (P4_s3_reg[WIDTH-1] & Cin_s3_reg);

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        S_out    <= 0;
        Cout_out <= 0;
    end else begin
        S_out    <= S_s4;
        Cout_out <= Cout_s4;
```

```
        end
    end

endmodule
```

Report



30-rcx_mcsta.nom.log

~/ee5311/tutorial_10/designs/koggestone/runs/auto_pnr/logs/signoff

```
67265
67266 ===========================================================================
67267 report_clock_skew
67268 ===========================================================================
67269 Clock core_clock
67270 Latency      CRPR       Skew
67271 _365_/CLK ^
67272    0.61
67273 _303_/CLK ^
67274    0.63      0.00      -0.02
67275
67276 skew_report_end
67277 summary_report
67278
67279 ===========================================================================
67280 report_tns
67281 ===========================================================================
67282 tns 0.00
67283
67284 ===========================================================================
67285 report_wns
67286 ===========================================================================
67287 wns 0.00
67288
67289 ===========================================================================
67290 report_worst_slack -max (Setup)
67291 ===========================================================================
67292 worst slack 2.68
67293
67294 ===========================================================================
67295 report_worst_slack -min (Hold)
67296 ===========================================================================
67297 worst slack 0.22
67298 summary_report_end
67299 check_nonpropagated_clocks
67300 check_nonpropagated_clocks_end
67301 [WARNING] Did not save OpenROAD database!
67302 Writing SDF files for all corners…
67303 Writing SDF for the Fastest corner to /openlane/designs/koggestone/runs/auto_pnr/results/routing/mca/process_corner_nom/koggestone.Fastest.sdf…
67304 Writing SDF for the Slowest corner to /openlane/designs/koggestone/runs/auto_pnr/results/routing/mca/process_corner_nom/koggestone.Slowest.sdf…
67305 Writing SDF for the Typical corner to /openlane/designs/koggestone/runs/auto_pnr/results/routing/mca/process_corner_nom/koggestone.Typical.sdf…
67306 Writing timing models for all corners…
67307 Writing timing models for the Fastest corner to /openlane/designs/koggestone/runs/auto_pnr/results/routing/mca/process_corner_nom/koggestone.Fastest.lib…
67308 Writing timing models for the Slowest corner to /openlane/designs/koggestone/runs/auto_pnr/results/routing/mca/process_corner_nom/koggestone.Slowest.lib…
67309 Writing timing models for the Typical corner to /openlane/designs/koggestone/runs/auto_pnr/results/routing/mca/process_corner_nom/koggestone.Typical.lib…
```

$$T_{\min} = \text{Target Period} - \text{Worst Setup Slack}$$

$$T_{\min} = 5.0 \text{ ns} - 2.68 \text{ ns} = \mathbf{2.32 \text{ ns}}$$

$$F_{\max} = \frac{1}{T_{\min}}$$

$$F_{\max} = \frac{1}{2.32 \text{ ns}} \approx \mathbf{431.03 \text{ MHz}}$$